

Pharmacotherapy for bronchiectasis.

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Bronchiectasis is a chronic lung disease and clinically is characterised by chronic cough, sputum production and susceptibility to lower respiratory tract infections. Management of bronchiectasis involves investigation to identify treatable causes and then focusing on prevention of exacerbations by regular physiotherapy and the appropriate use of antibiotics for infective exacerbations. There are few good quality, randomised, controlled trials in bronchiectasis and much of the treatment is based on expert opinion and extrapolation from the management of other respiratory conditions. This article discusses the management of bronchiectasis with particular reference to the pathophysiological processes involved in the condition. It then looks at the available evidence supporting such a management plan.

Secondary Immunodeficiency and Non-cystic Fibrosis Bronchiectasis.

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Bronchiectasis is a chronic respiratory disease characterized by abnormal dilation of the bronchi that causes cough, sputum, and recurrent infections. As it may be associated with various respiratory or systemic diseases, a critical aspect of managing bronchiectasis is to identify the underlying cause. Immunodeficiency is a rare but important cause of bronchiectasis, and its treatability is a significant trait for bronchiectasis management. While primary immunodeficiencies in bronchiectasis are well recognized, secondary immunodeficiencies remain under-reported and under-researched. Secondary immunodeficiencies may result from various diseases and conditions, such as hematologic malignancies, human immunodeficiency virus infection, renal transplantation, or the use of immunosuppressive drugs, and may contribute to the occurrence of bronchiectasis. Recurrent pulmonary and/or extrapulmonary infections in bronchiectasis may indicate the presence of secondary immunodeficiency in patients with these underlying conditions. For treatment, examining the underlying condition, managing bronchiectasis adequately, and prophylactic antibiotics (e.g., macrolide) and/or supplementary immunoglobulin G therapy may provide potential benefits. Considering the projected increase in the prevalence of secondary immunodeficiencies and bronchiectasis, future guidelines and research on the diagnosis and optimized treatment are needed.

Mechanisms of immune dysfunction and bacterial persistence in non-cystic fibrosis bronchiectasis.

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Bronchiectasis is a chronic inflammatory lung disease. The underlying cause is not identified in the majority of patients, but bronchiectasis is associated with a number of severe infections, immunodeficiencies and autoimmune disorders. Regardless of the underlying cause, the disease is characterised by a vicious cycle of bacterial colonisation, airway inflammation and airway structural damage. Inflammation in bronchiectasis is predominantly neutrophil driven. Neutrophils migrate to the airway under the action of pro-inflammatory cytokines such as interleukin-8, tumour necrosis factor- α and interleukin-1 β , all of which are increased in the airway of patients with bronchiectasis. Bacterial infection persists in the airway despite large numbers of neutrophils that would be expected to

phagocytose and kill pathogens under normal circumstances. Evidence suggests that neutrophils are disabled by multiple mechanisms including cleavage of phagocytic receptors by neutrophil elastase and inhibition of phagocytosis by neutrophil peptides. Complement activation is impaired and neutrophil elastase may cleave activated complement from pathogens preventing effective opsonisation. Organisms also evade clearance by adapting to chronic infection. The formation of biofilms, reduced motility and the down-regulation of virulence factors are among the strategies used to subvert innate immune mechanisms. Greater understanding of the mechanisms underlying chronic colonisation in bronchiectasis will assist in the development of new treatments for this important disease.

Management of bronchiectasis in adults.

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Once neglected in research and underappreciated in practice, there is renewed interest in bronchiectasis unrelated to cystic fibrosis. Bronchiectasis is a chronic lung disease characterised by chronic cough, sputum production and recurrent pulmonary exacerbations. It is diagnosed radiologically on high resolution computed tomography chest scan by bronchial dilatation (wider than the accompanying artery). The causes of bronchiectasis are diverse and include previous respiratory tract infections, chronic obstructive pulmonary disease, asthma, immunodeficiency and connective tissue diseases. A large proportion of cases are idiopathic, reflecting our incomplete understanding of disease pathogenesis. Progress in the evidence base is reflected in the 2017 European management guidelines and the 2015 update to the Australian guidelines. Effective airway clearance remains the cornerstone of bronchiectasis management. This should be personalised and reviewed regularly by a respiratory physiotherapist. There is now robust evidence for the long term use of oral macrolide antibiotics in selected patients to reduce exacerbation frequency. The routine use of long term inhaled corticosteroids and/or long-acting bronchodilators should be avoided, unless concomitant chronic obstructive pulmonary disease or asthma exists. The evidence for nebulised agents including hypertonic saline, mannitol and antibiotics is evolving; however, access is challenging outside tertiary clinics, and nebulising equipment is required. Smokers should be supported to quit. All patients should receive influenza and pneumococcal vaccination. Patients with impaired exercise capacity should attend pulmonary rehabilitation. There is an important minority of patients for whom aetiology-specific treatment exists. The prevalence of bronchiectasis is increasing worldwide; however, the burden of disease within Australia is not well defined. To this end, the Australian Bronchiectasis Registry began recruitment in 2016 and is interoperable with the European and United States bronchiectasis registries to enable collaborative research. The recent addition of a bronchiectasis diagnosis-related group to the Australian Refined Diagnostic Related Group classification system will allow definition of the disease burden within the Australian hospital system.

Bronchiectasis Development in a Male Patient With a History of Tuberculosis: A Case Report.

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Bronchiectasis is a chronic respiratory disease characterized by a syndrome of productive cough and recurrent respiratory infections due to permanent dilatation of the bronchi. In this case, we discuss a

32-year-old male patient with a history of tuberculosis (TB) from a rural area of Wardha, Maharashtra. The case discusses the diagnostic modalities confirming the diagnosis, sputum investigations, and imaging studies like chest X-ray, high-resolution computed tomography (HRCT), pulmonary function test (PFT), and bronchoscopy. This case underscores the importance of early recognition and management of bronchiectasis in patients with a history of pulmonary TB. Chronic inflammation and necrosis from the initial TB infection likely contributed to impaired mucociliary clearance and bronchial dilation, creating a conducive environment for bacterial colonization and recurrent infections. This case highlights the need for long-term follow-up and potential interventions to manage chronic respiratory symptoms in post-TB patients.

SAFE-ROCK: A Phase I Trial of an Oral Application of the ROCK Inhibitor Fasudil to Assess Bioavailability, Safety, and Tolerability in Healthy Participants.

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The intravenous (IV) formulation of Rho-kinase (ROCK) inhibitor fasudil has been approved for the treatment of subarachnoid haemorrhage since 1995. Additionally, fasudil has shown promising preclinical results for various chronic diseases, including neurodegenerative diseases such as amyotrophic lateral sclerosis, Parkinson's disease, and dementia, in which long-term intravenous (IV) administration might not be suitable. The objective of this study was to assess the absolute bioavailability of oral, in comparison to IV, application of the approved formulation of fasudil (ERIL®) and to evaluate the safety and tolerability of the oral application of fasudil. This was a phase I, single-center, open-label, randomized, two period cross-over clinical trial in healthy women and men. By applying a cross-over design, each subject served as their own control. Two treatments were investigated, separated by a wash out phase of at least 3 days. Oral fasudil was administered once on day 1 to assess pharmacokinetics and three times on day 2, at an interval of 8 ± 1 h, to assess safety and gastrointestinal tolerability. For pharmacokinetics of IV fasudil, it was administered once on day 1. Plasma profiles of fasudil and its active metabolite hydroxyfasudil after oral or IV administration were measured by liquid chromatography electrospray tandem mass spectrometry. Tolerability was assessed as proportion of subjects without significant drug intolerance, and safety was assessed by the proportion of subjects without clinical or laboratory treatment-associated serious adverse events. Gastrointestinal safety was assessed by applying the gastrointestinal symptom rating scale (GSRS). Fourteen subjects aged 30-70 years were included in this trial. After oral administration, fasudil concentrations in blood were mostly very low [1.4 g/L; coefficient of variation (CV) 41.0%]. After IV application, the peak concentration was 100.6 µg/L (CV 74.2%); however, a high variance in peak concentrations were assessed for both treatments. The maximal concentrations of hydroxyfasudil in blood were similar after oral and IV treatment [111.6 µg/L (CV 24.1%) and 108.4 µg/L (CV 19.7%), respectively]. Exposure of hydroxyfasudil (assessed as AUC_{0-tz}) differed between both treatments, with 449 µg × h/L after IV treatment and 309 µg × h/L after oral treatment. Therefore, the absolute bioavailability of hydroxyfasudil after the oral treatment was approximately 69% of the IV treatment. No serious adverse events (SAEs) occurred during this trial, and good tolerability of oral fasudil (90 mg/day) was documented. Oral fasudil was generally well tolerated in the studied population, and no safety concerns were identified. However, systemic bioavailability of oral hydroxyfasudil corresponded to 69%, and dose adjustments need to be considered. The results presented here lay grounds for future trials of fasudil in chronic diseases, which require an oral long-term application. This trial was registered with EudraCT (no. 2019-001805-26).

Protocol for a randomized, placebo-controlled, double-blind phase IIa study of the safety, tolerability, and symptomatic efficacy of the ROCK-inhibitor Fasudil in patients with Parkinson's disease (ROCK-PD).

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The Rho-kinase (ROCK) inhibitor Fasudil has shown symptomatic and disease-modifying effects in Parkinson's disease (PD) models in vitro and in vivo. In Japan, Fasudil has been approved for the treatment of subarachnoid haemorrhage since 1995 and shows a favourable safety profile. To investigate the safety, tolerability, and symptomatic efficacy of ROCK-inhibitor Fasudil in comparison to placebo in a randomized, national, multicenter, double-blind phase IIa study in patients with PD. We plan to include 75 patients with at least 'probable' PD (MDS criteria), Hoehn and Yahr stages 1-3, and age 30-80 years in 13 German study sites. Patients must be non-fluctuating and their response to PD medication must have been stable for 6 weeks. Patients will be randomly allocated to treatment with the oral investigational medicinal product (IMP) containing either Fasudil in two dosages, or placebo, for a total of 22 days. As primary analysis, non-inferiority of low/high dose of Fasudil on the combined endpoint consisting of occurrence of intolerance and/or treatment-related serious adverse events (SAEs) over 22 days will be assessed in a sequential order, starting with the lower dose. Secondary endpoints will include tolerability alone over 22 days and occurrence of treatment-related SAEs (SARs) over 22 and 50 days and will be compared on group level. Additional secondary endpoints include efficacy on motor and non-motor symptoms, measured on established scales, and will be assessed at several timepoints. Biomaterial will be collected to determine pharmacokinetics of Fasudil and its active metabolite, and to evaluate biomarkers of neurodegeneration. After positive evaluation by the competent authority and the ethics committee, patient recruitment started in the 3rd quarter of 2023. ROCK-PD is registered with Eudra-CT (2021-003879-34) and clinicaltrials.gov (NCT05931575). Results of this trial can pave way for conducting extended-duration studies assessing both symptomatic efficacy and disease-modifying properties of Fasudil.

Pharmacotherapy in SAH: Clinical Trial Lessons.

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Subarachnoid Haemorrhage (SAH) is a medical emergency with potentially devastating outcomes. It is without doubt that over the past decades, there has been a radical change in the approach towards patients with SAH, both in terms of the surgical as well as of the pharmacological treatments offered. The present review aims to outline the principal data regarding the best practice in the pharmacotherapy of SAH, as well as to sum up the emerging evidence from the latest clinical trials. To date, nimodipine is the only evidence-based treatment of vasospasm. However, extensive research is currently underway to identify novel substances with magnesium sulphate, cilostazol, clazosentan and fasudil, demonstrating promising results. Antifibrinolytic therapy could help reduce mortality, and anticoagulants, in spite of their associated hazards, could actually reduce the incidence of delayed cerebral ischemia. The effectiveness of triple-H therapy has been challenged, yet evidence on the optimal regimen is still pending. Statins may benefit some patients by reducing the incidence of vasospasm and delayed ischemic events. As several clinical trials are underway, it is expected that in the years to come, more therapeutic options will be added to the attending physician's armamentarium.

Safety, tolerability, and efficacy of fasudil in amyotrophic lateral sclerosis (ROCK-ALS): a phase 2, randomised, double-blind, placebo-controlled trial.

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Fasudil is a small molecule inhibitor of Rho-associated kinase (ROCK) and is approved for the treatment of subarachnoid haemorrhage. In preclinical studies, fasudil has been shown to attenuate neurodegeneration, modulate neuroinflammation, and foster axonal regeneration. We aimed to investigate the safety, tolerability, and efficacy of fasudil in patients with amyotrophic lateral sclerosis. ROCK-ALS was a phase 2, randomised, double-blind, placebo-controlled trial conducted at 19 amyotrophic lateral sclerosis centres in Germany, France, and Switzerland. Individuals (aged 18-80 years) with at least probable amyotrophic lateral sclerosis (as per the revised El Escorial criteria), a disease duration of 6-24 months, and a slow vital capacity greater than 65% of predicted normal were eligible for inclusion. Patients were randomly assigned (1:1:1) to receive 30 mg (15 mg twice daily) or 60 mg (30 mg twice daily) fasudil or matched placebo intravenously for 20 days over a 4-week period. Follow-up assessments were performed at 45, 90, and 180 days after treatment initiation. The co-primary endpoints were safety until day 180 (defined as the proportion without drug-related serious adverse events) and tolerability during the treatment period (defined as the proportion who did not discontinue treatment due to suspected drug-related adverse events). The primary analyses were carried out in the intention-to-treat population, which included all participants who entered the treatment phase. This trial is registered at ClinicalTrials.gov (NCT03792490) and Eudra-CT (2017-003676-31) and is now completed. Between Feb 20, 2019, and April 20, 2022, 120 participants were enrolled and randomised; two individuals assigned fasudil 30 mg withdrew consent before the baseline visit. Thus, the intention-to-treat population comprised 35 in the fasudil 30 mg group, 39 in the fasudil 60 mg group, and 44 in the placebo group. The estimated proportion without a drug-related serious adverse event was 1.00 (95% CI 0.91 to 1.00) with placebo, 1.00 (0.89 to 1.00) with fasudil 30 mg, and 1.00 (0.90 to 1.00) with fasudil 60 mg; the difference in proportions was 0.00 (95% CI -0.11 to 0.10; $p>0.99$) for fasudil 30 mg versus placebo and 0.00 (-0.10 to 0.10; $p>0.99$) for fasudil 60 mg versus placebo. Treatment tolerability (the estimated proportion who did not discontinue) was 0.93 (95% CI 0.81 to 0.99) with placebo, 1.00 (0.90 to 1.00) with fasudil 30 mg, and 0.90 (0.76 to 0.97) with fasudil 60 mg; the difference in proportions was 0.07 (95% CI -0.05 to 0.20; $p=0.25$) for fasudil 30 mg versus placebo, and -0.03 (-0.18 to 0.10; $p=0.70$) for fasudil 60 mg versus placebo. Eight deaths occurred: two in the placebo group, four in the fasudil 30 mg group, and two in the fasudil 60 mg group. The most common serious adverse events were respiratory failure (seven events), gastrostomy (five events), pneumonia (four events), and dysphagia (four events). No serious adverse events or deaths were attributed to study treatment. Adverse events, which were mainly related to disease progression, occurred in 139 participants in the placebo group, 108 in the fasudil 30 mg group, and 105 in the fasudil 60 mg group. Fasudil was well tolerated and safe in people with amyotrophic lateral sclerosis. The effect of fasudil on efficacy outcomes should be explored in larger clinical trials with a longer treatment duration, oral administration, and potentially higher dose of the trial drug. Framework of the E-Rare Joint Transnational Call 2016 "Clinical research for new therapeutic uses of already existing molecules (repurposing) in rare diseases".

Combination therapy of fasudil hydrochloride and ozagrel sodium for cerebral vasospasm following aneurysmal subarachnoid hemorrhage.

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Fasudil hydrochloride is a new type of intracellular calcium antagonist, different from the calcium entry blockers that are commonly employed for clinical use. Since September 1995, the combination of fasudil hydrochloride and ozagrel sodium, an inhibitor of thromboxane A2 synthesis, has been used to treat 60 patients at risk of cerebral vasospasm after aneurysmal subarachnoid hemorrhage. The effectiveness of this combination therapy was investigated by comparison with the outcome of 57 patients previously treated with only ozagrel sodium. The combination therapy was significantly more effective ($p < 0.01$) in reducing the incidence of low density areas on computed tomography scans, and reduced, but not significantly, the occurrence of symptomatic vasospasm. The combination therapy of fasudil hydrochloride and ozagrel sodium has superior effectiveness over only ozagrel sodium in treating patients at risk of vasospasm after aneurysmal subarachnoid hemorrhage.
